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Control method for gas turbine combustor and control device for gas turbine combustor

Abstract

A control method for a gas turbine combustor including an air-hole plate having a plurality of air holes formed therein, and a plurality of fuel nozzles corresponding to respective ones of the plurality of air holes. The gas turbine combustor is operable to combust a hydrogen fuel and a fuel other than the hydrogen fuel. The air holes include: first air holes having, among passageways from an entry end to an exit end, an inclined passageway extending in a direction inclined with respect to the central axis of the air-hole plate in a region including at least the exit end; and second air holes extending in parallel to the central axis. The fuel nozzles include first fuel nozzles corresponding to respective first air holes, and second fuel nozzles corresponding to respective second air holes. During hydrogen fuel firing, the hydrogen fuel is not supplied to the first fuel nozzles.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a control method for a gas turbine combustor and a control device for a gas turbine combustor.

(2) The present application claims priority based on Japanese Patent Application No. 2022-006878 filed in Japan on Jan. 20, 2022, the contents of which are incorporated herein by reference.

BACKGROUND ART

(3) A gas turbine combustor is known in which an air-hole plate is disposed between a fuel nozzle and a combustion chamber and which is configured to eject a fuel flow and an air flow formed on an outer peripheral side of the fuel flow into the combustion chamber inside air holes provided in the air-hole plate (for example, see PTL 1).

CITATION LIST

Patent Literature

(4) [PTL 1] Japanese Unexamined Patent Application Publication No. 2018-150912

SUMMARY OF INVENTION

Technical Problem

(5) For example, as in a gas turbine combustor disclosed in PTL 1, in a case where the air holes extend in a direction inclined with respect to a central axis of the air-hole plate, there is a concern that a region where a flow velocity is relatively small may occur locally in the air-hole plate or at a position relatively close to the air-hole plate. For example, in a case in which hydrogen having a relatively fast combustion speed is used as a fuel, the above-described region may be a starting point for an occurrence of backfire, and in a case in which the backfire occurs, there is a concern that a flame may continuously remain in the above-described region. In a case where the flame

continuously remains in the region, there is a concern that the combustor may be damaged.

(6) At least one embodiment of the present disclosure aims to suppress damage to a gas turbine combustor that combusts a hydrogen fuel and another fuel other than the hydrogen fuel in view of the above-described circumstances.

Solution to Problem

(7) (1) A control method for a gas turbine combustor according to at least one embodiment of the present disclosure is a control method for the gas turbine combustor that includes an air-hole plate in which a plurality of air holes are formed and a plurality of fuel nozzles corresponding to the plurality of air holes, respectively, and that combusts a hydrogen fuel and another fuel other than the hydrogen fuel, in which the plurality of air holes include a plurality of first air holes each having an inclined passageway that extends in a direction inclined with respect to a central axis of the air-hole plate in a region including at least an exit end in a passageway from an entry end to the exit end, and a plurality of second air holes that extend parallel to the central axis, and the plurality of fuel nozzles include a plurality of first fuel nozzles corresponding to the plurality of first air holes, respectively, and a plurality of second fuel nozzles corresponding to the plurality of second air holes, respectively, the control method including: supplying no hydrogen fuel to the plurality of first fuel nozzles when the hydrogen fuel is exclusively combusted.

(8) (2) A control device for a gas turbine combustor according to at least one embodiment of the present disclosure is a control device for controlling combustion in the gas turbine combustor that includes an air-hole plate in which a plurality of air holes are formed and a plurality of fuel nozzles corresponding to the plurality of air holes, respectively, and that combusts a hydrogen fuel and another fuel other than the hydrogen fuel, in which the plurality of air holes include a plurality of first air holes each having an inclined passageway that extends in a direction inclined with respect to a central axis of the air-hole plate in a region including at least an exit end in a passageway from an entry end to the exit end, and a plurality of second air holes that extend parallel to the central axis, and the plurality of fuel nozzles include a plurality of first fuel nozzles corresponding to the plurality of first air holes, respectively, and a plurality of second fuel nozzles corresponding to the plurality of second air holes, respectively, the control device including: a fuel flow rate regulating valve that regulates a flow rate of a fuel supplied to the plurality of first fuel nozzles, and a fuel flow rate control unit that controls the fuel flow rate regulating valve, in which the fuel flow rate control unit controls the fuel flow rate regulating valve such that no hydrogen fuel is supplied to the plurality of first fuel nozzles when the hydrogen fuel is exclusively combusted.

Advantageous Effects of Invention

(9) According to at least one embodiment of the present disclosure, it is possible to suppress damage to a gas turbine combustor that combusts a hydrogen fuel and another fuel other than the hydrogen fuel.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 shows a schematic configuration of a gas turbine including a gas turbine combustor according to some embodiments.

(2) FIG. 2A is a schematic partial sectional view showing a structure in a vicinity of a burner in a gas turbine combustor according to one embodiment provided in the gas turbine shown in FIG. 1.

(3) FIG. 2B is a schematic partial sectional view showing a structure in a vicinity of a burner in a gas turbine combustor according to another embodiment provided in the gas turbine shown in FIG. 1.

(4) FIG. 3A is a view of an air-hole plate of the burner according to the embodiment shown in FIG. 2A when viewed from an axial downstream side.

- (5) FIG. 3B is a view of an air-hole plate of the burner according to the other embodiment shown in FIG. 2B when viewed from the axial downstream side.
- (6) FIG. 4A is a view showing an example of a first air hole in an air hole.
- (7) FIG. 4B is a view showing another example of the first air hole in the air hole.
- (8) FIG. 5A is a view for describing control of a combustion injection ratio based on a hydrogen mixed combustion ratio in the gas turbine combustor according to the embodiment shown in FIGS. 2A and 3A.
- (9) FIG. 5B is a view for describing control of a combustion injection ratio based on a hydrogen mixed combustion ratio in the gas turbine combustor according to the other embodiment shown in FIGS. 2B and 3B.
- (10) FIG. 6A is a graph showing a transition of a fuel ratio and the hydrogen mixed combustion ratio from a start of operation of the gas turbine provided with the gas turbine combustor according to the embodiment shown in FIGS. 2A and 3A to a transition to exclusive combustion of a hydrogen fuel.
- (11) FIG. 6B is a graph showing a transition of the fuel ratio and the hydrogen mixed combustion ratio from a start of operation of the gas turbine provided with the gas turbine combustor according to the other embodiment shown in FIGS. 2B and 3B to a transition to exclusive combustion of the hydrogen fuel.

DESCRIPTION OF EMBODIMENTS

- (12) Hereinafter, some embodiments of the present disclosure will be described with reference to the accompanying drawings. Dimensions, materials, shapes, relative arrangements, and the like of components described as embodiments or illustrated in the drawings are not intended to limit the scope of the present disclosure, but are merely explanatory examples.
- (13) For example, an expression representing a relative or absolute arrangement such as “in a certain direction”, “along a certain direction”, “parallel”, “orthogonal”, “center”, “concentric”, or “coaxial” does not strictly represent only such an arrangement, but also a tolerance or a state of being relatively displaced with an angle or a distance to the extent that the same function can be obtained.
- (14) For example, expressions such as “identical”, “equal”, and “homogeneous” indicating that things are in an equal state do not strictly represent only the equal state, but also a tolerance or a state where there is a difference to the extent that the same function can be obtained.
- (15) For example, an expression representing a shape such as a quadrangular shape or a cylindrical shape does not represent only a shape such as a quadrangular shape or a cylindrical shape in a geometrically strict sense, but also a shape including an uneven portion, a chamfered portion, and the like within a range in which the same effect can be obtained.
- (16) Meanwhile, the expressions “being provided with”, “comprising”, “including”, or “having” one component are not exclusive expressions excluding the presence of other components.
- (17) A gas turbine combustor according to some embodiments of the present disclosure will be described with reference to FIGS. 1, and 2A and 2B.
- (18) FIG. 1 shows a schematic configuration of a gas turbine including a gas turbine combustor according to some embodiments of the present disclosure.
- (19) A gas turbine 1 shown in FIG. 1 includes an air compressor 110, a gas turbine combustor 100, and a turbine 180.
- (20) The gas turbine combustor 100 according to some embodiments includes a combustor liner (inner cylinder) 153, a liner flow sleeve (outer cylinder) 154, a transition piece 152, a transition piece flow sleeve 150, a burner 200, a fuel system 300, and a control device 10 (see FIG. 2A and FIG. 2B). In FIG. 1, the burner 200 and the fuel system 300 are shown in a simplified manner, and only one fuel header 230 and only one fuel supply pipe 305, which will be described below, are shown. The burner 200 and the fuel system 300 will be described later.
- (21) In the gas turbine 1 shown in FIG. 1, the air compressor 110 is rotationally driven by the

turbine **180** to compress air (suctioned air) suctioned from the atmosphere through an intake portion (not shown) to generate high-pressure air (combustion air) **120**, and supplies the high-pressure air **120** to the gas turbine combustor **100**. The gas turbine combustor **100** combusts the high-pressure air **120** supplied from the air compressor **110** by mixing the high-pressure air **120** with a fuel supplied from the fuel system **300** to generate a high-temperature combustion gas **170**, and supplies the high-temperature combustion gas **170** to the turbine **180**.

(22) That is, in the gas turbine **1** shown in FIG. **1**, the high-pressure air **120** which is combustion air discharged from the air compressor **110** is introduced into a casing **140** from a diffuser **130**, and flows into a flow path formed in a gap between the transition piece flow sleeve **150** and the transition piece **152** disposed inside the transition piece flow sleeve **150** from an air introduction hole **151** provided in the transition piece flow sleeve **150** of the gas turbine combustor **100**.

(23) Thereafter, the high-pressure air **120** flowing into the flow path formed in the gap flows into a flow path formed in a gap between the combustor liner **153** of the gas turbine combustor **100** and the liner flow sleeve **154** disposed concentrically with the combustor liner **153** on an outer peripheral side of the combustor liner **153**. Thereafter, a flow of the high-pressure air **120** is reversed and introduced from the fuel system **300**. The high-pressure air **120** is mixed with a fuel injected from a plurality of fuel nozzles **210** forming a cluster nozzle, and is combusted in a combustion chamber **160** inside the combustor liner **153** to form a flame **156** and to generate a high-temperature and high-pressure combustion gas **170**.

(24) In this way, the high-temperature and high-pressure combustion gas **170** generated in the gas turbine combustor **100** flows down in the transition piece **152**, and is introduced into the turbine **180**.

(25) In the turbine **180** forming the gas turbine **1**, a work amount generated when the high-temperature and high-pressure combustion gas **170** introduced into the turbine **180** is subjected to adiabatic expansion is converted into a shaft rotational force by the turbine **180**. In this manner, an output is obtained from a generator **190** by driving the generator **190** connected to the turbine **180** by a turbine shaft.

(26) The air compressor **110** and the generator **190** which form the gas turbine **1** are connected to the turbine **180** by the turbine shaft. However, the air compressor **110**, the turbine **180**, and the generator **190** may not have a configuration in which the turbine shaft has a single shaft, and may have a configuration in which the turbine shaft has two or more shafts.

(27) In addition, the gas turbine which is generally widely used in a thermal power plant has a configuration in which a plurality of the gas turbine combustors are radially arranged with respect to the turbine shaft.

(28) FIG. **2A** is a schematic partial sectional view showing a structure in a vicinity of the burner **200** in the gas turbine combustor **100** according to one embodiment provided in the gas turbine **1** shown in FIG. **1**.

(29) FIG. **2B** is a schematic partial sectional view showing a structure in the vicinity of the burner **200** in the gas turbine combustor **100** according to another embodiment provided in the gas turbine **1** shown in FIG. **1**.

(30) FIG. **3A** is a view of an air-hole plate **25** of the burner **200** according to the embodiment shown in FIG. **2A** when viewed from an axial downstream side.

(31) FIG. **3B** is a view of the air-hole plate **25** of the burner **200** according to the other embodiment shown in FIG. **2B** when viewed from the axial downstream side.

(32) In the following description, a direction along a central axis **AXc** of the gas turbine combustor **100** will be referred to as an axial direction of the gas turbine combustor **100**, or will be also simply referred to as an axial direction. A direction in which the combustion gas **170** flows along the axial direction will be referred to as the axial downstream side, or will be simply referred to as a downstream side. A direction opposite to the direction in which the combustion gas **170** flows will be referred to as an axial upstream side, or will be simply referred to as an upstream side.

- (33) In the gas turbine combustor **100** according to some embodiments, the central axis AXc of the gas turbine combustor **100** is, for example, a central axis of the combustor liner **153** having a cylindrical shape. In the gas turbine combustor **100** according to some embodiments, the central axis AXc of the gas turbine combustor **100** coincides with a central axis AXp of the air-hole plate **25**.
- (34) (About Details of Gas Turbine Combustor **100**)
- (35) The gas turbine combustor **100** according to some embodiments is a gas turbine combustor capable of combusting a hydrogen fuel and another fuel other than the hydrogen fuel. In the gas turbine combustor **100** according to some embodiments, a natural gas fuel is combusted as the other fuel. In the gas turbine combustor **100** according to some embodiments, the hydrogen fuel can be exclusively combusted, the natural gas fuel can be exclusively combusted, and the hydrogen fuel and the natural gas fuel can be mixed-combusted.
- (36) As shown in FIGS. 2A and 2B, in the gas turbine combustor **100** according to some embodiments, the burner **200** is disposed to be orthogonal to the central axis AXc (central axis of the combustor liner **153**) of the gas turbine combustor **100**, and is provided at an end portion of the combustor liner **153** on the axial upstream side. In the gas turbine combustor **100** according to some embodiments, the burner **200** includes the fuel header **230**, the plurality of fuel nozzles **210**, and the air-hole plate **25**.
- (37) The gas turbine combustor **100** according to some embodiments is a combustor of a type called a cluster combustor. In the gas turbine combustor **100** according to some embodiments, a plurality of air holes **250** are formed in the air-hole plate **25**. Each of the plurality of fuel nozzles **210** is disposed to correspond to each of the plurality of air holes **250** formed in the air-hole plate **25** disposed close to an axial downstream side of the fuel nozzles **210**.
- (38) A tip of each of the fuel nozzles **210** need not be inserted into each of the air holes **250** as shown in FIGS. 2A and 2B, or may be inserted into each of the air holes **250**.
- (39) As shown in FIGS. 2A, 2B, 3A, and 3B, in the gas turbine combustor **100** according to some embodiments, the burner **200** is a so-called multi-burner including one central burner **211** disposed coaxially with the combustor liner **153** at a center and a plurality of (six in the present embodiment) outer burners **212** disposed around the central burner **211**. The central burner **211** and the outer burners **212** are each divided into a plurality of (three in the present embodiment) concentric annular rows. In the following description, the plurality of annular rows of the central burner **211** and the outer burners **212** are appropriately referred to as a first row, a second row, and a third row, respectively, from an inner peripheral side toward an outer peripheral side.
- (40) (Central Burner **211**)
- (41) In the gas turbine combustor **100** according to some embodiments, the central burner **211** includes the fuel header **230**, the plurality of fuel nozzles **210**, and the plurality of air holes **250** formed in the air-hole plate **25**. The central burner **211** is supported by the fuel header **230**. The fuel nozzle **210** of the central burner **211** is disposed concentrically in the first to third rows of the central burner **211** and is provided over an entire circumference of each row (is annularly disposed). The fuel nozzle **210** of the central burner **211** injects a fuel supplied from the fuel system **300** toward the air holes **250** formed in the air-hole plate **25**.
- (42) In the gas turbine combustor **100** according to some embodiments, the fuel header **230** of the central burner **211** includes an inner peripheral fuel header **231** and an outer peripheral fuel header **232** provided to surround a radial outer side of the inner peripheral fuel header **231**.
- (43) In the gas turbine combustor **100** according to some embodiments, the plurality of fuel nozzles **210** of the central burner **211** include an inner peripheral fuel nozzle **210i** connected to the inner peripheral fuel header **231** and an outer peripheral fuel nozzle **210o** connected to the outer peripheral fuel header **232**.
- (44) In the gas turbine combustor **100** according to some embodiments, the inner peripheral fuel nozzle **210i** of the central burner **211** corresponds to the fuel nozzle **210** of the first row, and the

outer peripheral fuel nozzles **210o** of the central burner **211** correspond to the fuel nozzles **210** of the second row and the third row.

(45) In the gas turbine combustor **100** according to some embodiments, the air-hole plate **25** includes a plurality of inner peripheral air holes **250i** corresponding to a plurality of the inner peripheral fuel nozzles **210i** of the central burner **211**, respectively, and a plurality of outer peripheral air holes **250o** corresponding to a plurality of the outer peripheral fuel nozzles **210o** of the central burner **211**, respectively.

(46) In the gas turbine combustor **100** according to some embodiments, the first row of the central burner **211** is composed of the inner peripheral fuel header **231**, the inner peripheral fuel nozzle **210i**, and the inner peripheral air hole **250i**, and the second and third rows are composed of the outer peripheral fuel header **232**, the outer peripheral fuel nozzle **210o**, and the outer peripheral air hole **250o**.

(47) (Outer Burner **212**)

(48) In the gas turbine combustor **100** according to some embodiments, the outer burner **212** includes the fuel header **230**, the plurality of fuel nozzles **210**, and the plurality of air holes **250** formed in the air-hole plate **25**. The outer burner **212** is supported by the fuel header **230**. The fuel nozzle **210** of the outer burner **212** is disposed concentrically in the first to third rows of the outer burner **212** and is provided over an entire circumference of each row (is annularly disposed). The fuel nozzle **210** of the outer burner **212** injects the fuel supplied from the fuel system **300** toward the air holes **250** formed in the air-hole plate **25**.

(49) In the gas turbine combustor **100** according to some embodiments, the fuel header **230** of the outer burner **212** includes the inner peripheral fuel header **231** and the outer peripheral fuel header **232** provided to surround the radial outer side of the inner peripheral fuel header **231**.

(50) In the gas turbine combustor **100** according to some embodiments, the plurality of fuel nozzles **210** of the outer burner **212** include the inner peripheral fuel nozzle **210i** connected to the inner peripheral fuel header **231** and the outer peripheral fuel nozzle **210o** connected to the outer peripheral fuel header **232**.

(51) In the gas turbine combustor **100** according to some embodiments, the inner peripheral fuel nozzle **210i** of the outer burner **212** corresponds to the fuel nozzle **210** of the first row, and the outer peripheral fuel nozzles **210o** of the outer burner **212** correspond to the fuel nozzles **210** of the second row and the third row.

(52) In the gas turbine combustor **100** according to some embodiments, the air-hole plate **25** includes the plurality of inner peripheral air holes **250i** corresponding to the plurality of inner peripheral fuel nozzles **210i** of the outer burner **212**, respectively, and the plurality of outer peripheral air holes **250o** corresponding to the plurality of outer peripheral fuel nozzles **210o** of the outer burner **212**, respectively.

(53) In the gas turbine combustor **100** according to some embodiments, the first row of the outer burner **212** is composed of the inner peripheral fuel header **231**, the inner peripheral fuel nozzle **210i**, and the inner peripheral air hole **250i**, and the second and third rows are composed of the outer peripheral fuel header **232**, the outer peripheral fuel nozzle **210o**, and the outer peripheral air hole **250o**.

(54) FIG. 4A is a view showing an example of a first air hole **251** in the air hole **250**.

(55) FIG. 4B is a view showing another example of the first air hole **251** in the air hole **250**.

(56) As shown in FIGS. 4A and 4B, in the gas turbine combustor **100** according to some embodiments, the plurality of air holes **250** include a plurality of the first air holes **251** each having an inclined passageway **256** that extends in a direction inclined with respect to the central axis AXp of the air-hole plate **25** in a region including at least an exit end **250b** in a passageway **255** from an entry end **250a** to the exit end **250b**, and a plurality of second air holes **252** that extend parallel to the central axis AXp of the air-hole plate **25**.

(57) The inclined passageway **256** in the first air hole **251** may be only a part of the passageway

255 on an exit end **250b** side (axial downstream side) as shown in FIG. 4A, and the passageway **255** on an entry end **250a** side (axial upstream side) may extend parallel to the central axis AXp of the air-hole plate **25**.

(58) In addition, as shown in FIG. 4B, the inclined passageway **256** in the first air hole **251** may extend from the entry end **250a** to the exit end **250b**.

(59) In the gas turbine combustor **100** according to some embodiments, the fuel nozzle **210** corresponding to the first air hole **251** is also referred to as a first fuel nozzle **21**, and the fuel nozzle **210** corresponding to the second air hole **252** is also referred to as a second fuel nozzle **22**.

(60) (About Air Hole **250** of Burner **200** According to Embodiment)

(61) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, the inner peripheral air hole **250i** corresponding to the fuel nozzle **210** of the first row of the central burner **211**, that is, the inner peripheral fuel nozzle **210i**, is the first air hole **251**.

(62) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, the inner peripheral air holes **250i** of the central burner **211** constitute a first air hole group G1 in which the plurality of first air holes **251** are disposed adjacent to each other.

(63) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, the fuel nozzle **210** of the first row of the central burner **211**, that is, the inner peripheral fuel nozzle **210i**, is the first fuel nozzle **21**.

(64) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, the outer peripheral air holes **250o** corresponding to the fuel nozzles **210** of the second row and the third row of the central burner **211**, that is, the outer peripheral fuel nozzles **210o**, are the second air holes **252**.

(65) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, the outer peripheral air holes **250o** of the central burner **211** constitute a second air hole group G2 in which the plurality of second air holes **252** are disposed adjacent to each other.

(66) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, the fuel nozzles **210** of the second row and the third row of the central burner **211**, that is, the outer peripheral fuel nozzles **210o**, are the second fuel nozzles **22**.

(67) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, the inner peripheral air hole **250i** and the outer peripheral air holes **250o** corresponding to the fuel nozzles **210** of the first row and the second and third rows of the outer burner **212**, that is, the inner peripheral fuel nozzle **210i** and the outer peripheral fuel nozzles **210o**, respectively, are the second air holes **252**.

(68) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, the inner peripheral air hole **250i** and the outer peripheral air hole **250o** of the outer burner **212** constitute the second air hole group G2.

(69) In the burner **200** according to the embodiment shown in FIG. 2A and FIG. 3A, all the fuel nozzles **210** of the outer burner **212** are the second fuel nozzles **22**.

(70) (About Air Hole **250** of Burner **200** According to Other Embodiment)

(71) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the inner peripheral air hole **250i** corresponding to the fuel nozzle **210** of the first row of the central burner **211**, that is, the inner peripheral fuel nozzle **210i**, is the first air hole **251**.

(72) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the inner peripheral air holes **250i** of the central burner **211** constitute the first air hole group G1.

(73) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the fuel nozzle **210** of the first row of the central burner **211**, that is, the inner peripheral fuel nozzle **210i**, is the first fuel nozzle **21**.

(74) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the outer peripheral air holes **250o** corresponding to the fuel nozzles **210** of the second row and the third row of the central burner **211**, that is, the outer peripheral fuel nozzles **210o**, are the second air holes

252.

(75) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the outer peripheral air holes **250o** of the central burner **211** constitute the second air hole group G2.

(76) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the fuel nozzles **210** of the second row and the third row of the central burner **211**, that is, the outer peripheral fuel nozzles **210o**, are the second fuel nozzles **22**.

(77) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the inner peripheral air hole **250i** corresponding to the fuel nozzle **210** of the first row of the outer burner **212**, that is, the inner peripheral fuel nozzle **210i**, is the first air hole **251**.

(78) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the inner peripheral air holes **250i** of the outer burner **212** constitute the first air hole group G1.

(79) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the fuel nozzle **210** of the first row of the outer burner **212**, that is, the inner peripheral fuel nozzle **210i**, is the first fuel nozzle **21**.

(80) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the outer peripheral air holes **250o** corresponding to the fuel nozzles **210** of the second row and the third row of the outer burner **212**, that is, the outer peripheral fuel nozzles **210o**, are the second air holes **252**.

(81) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the outer peripheral air holes **250o** of the outer burner **212** constitute the second air hole group G2.

(82) In the burner **200** according to the other embodiment shown in FIG. 2B and FIG. 3B, the fuel nozzles **210** of the second row and the third row of the outer burner **212**, that is, the outer peripheral fuel nozzles **210o**, are the second fuel nozzles **22**.

(83) That is, in the burner **200** according to the embodiment shown in FIGS. 2A and 3A, the inner peripheral air hole **250i** of the outer burner **212** is the second air hole **252**, but in the burner **200** according to the other embodiment shown in FIGS. 2B and 3B, the inner peripheral air hole **250i** of the outer burner **212** is the first air hole **251**. The burner **200** according to the embodiment shown in FIGS. 2A and 3A and the burner **200** according to the other embodiment shown in FIGS. 2B and 3B have the same configuration except that a type of the inner peripheral air hole **250i** of the outer burner **212** is different and a type of the inner peripheral fuel nozzle **210i** of the outer burner **212** is different.

(84) (Fuel System **300**)

(85) In the gas turbine combustor **100** according to some embodiments, the fuel system **300** includes a hydrogen fuel pipe **301** that is a fuel pipe of the hydrogen fuel, and a natural gas fuel pipe **302** that is a fuel pipe of the natural gas fuel. In the gas turbine combustor **100** according to some embodiments, the fuel system **300** includes a mixing device (mixer) **307** for generating a mixed fuel of the hydrogen fuel and the natural gas fuel, and the fuel supply pipe **305** for supplying the mixed fuel from the mixing device **307** or the hydrogen fuel or the natural gas fuel supplied through the mixing device **307** to each fuel header **230**.

(86) In the gas turbine combustor **100** according to some embodiments, the fuel system **300** includes a plurality of fuel flow rate regulating valves **310** and **320** for regulating a flow rate of the fuel.

(87) In the gas turbine combustor **100** according to some embodiments, the fuel flow rate regulating valve **310** includes a hydrogen flow rate regulating valve **311** for regulating a flow rate of the hydrogen fuel supplied to the mixing device **307** via the hydrogen fuel pipe **301**, and a natural gas flow rate regulating valve **312** for regulating a flow rate of the natural gas fuel supplied to the mixing device **307** via the natural gas fuel pipe **302**.

(88) In the gas turbine combustor **100** according to some embodiments, the hydrogen flow rate regulating valve **311** and the natural gas flow rate regulating valve **312** are each provided with an actuator (not shown) for changing a valve opening degree. In the gas turbine combustor **100** according to some embodiments, a configuration is adopted so that a control signal for driving the

actuators is output from a fuel flow rate control unit **11** to be described below.

(89) In the gas turbine combustor **100** according to some embodiments, the fuel flow rate regulating valve **320** is a fuel flow rate regulating valve for regulating a flow rate of the fuel supplied to each fuel header **230**. The fuel flow rate regulating valve **320** includes a first fuel flow rate regulating valve **321** for regulating a flow rate of the fuel supplied to the inner peripheral fuel header **231** of the central burner **211**, a second fuel flow rate regulating valve **322** for regulating a flow rate of the fuel supplied to the outer peripheral fuel header **232** of the central burner **211**, a third fuel flow rate regulating valve **323** for regulating a flow rate of the fuel supplied to the inner peripheral fuel header **231** of the outer burner **212**, and a fourth fuel flow rate regulating valve **324** for regulating a flow rate of the fuel supplied to the outer peripheral fuel header **232** of the outer burner **212**.

(90) In the gas turbine combustor **100** according to some embodiments, the first fuel flow rate regulating valve **321**, the second fuel flow rate regulating valve **322**, the third fuel flow rate regulating valve **323**, and the fourth fuel flow rate regulating valve **324** are each provided with an actuator (not shown) for changing a valve opening degree.

(91) In the gas turbine combustor **100** according to some embodiments, a configuration is adopted so that a control signal for driving the actuators is output from a fuel flow rate control unit **11** to be described below.

(92) (Control Device **10**)

(93) In the gas turbine combustor **100** according to some embodiments, the control device **10** includes the plurality of fuel flow rate regulating valves **310** and **320**, and the fuel flow rate control unit **11** that controls the plurality of fuel flow rate regulating valves **310** and **320**.

(94) The fuel flow rate control unit **11** includes a processor **12** that performs various arithmetic processes, and a memory **13** that non-temporarily or temporarily stores various data processed by the processor **12**. The processor **12** is realized by a CPU, a GPU, an MPU, a DSP, various other arithmetic devices, or a combination thereof. The memory **13** is realized by a ROM, a RAM, a flash memory, or a combination thereof.

(95) (Problem Caused by Use of Hydrogen Fuel)

(96) As in the first air hole **251** in the gas turbine combustor **100** according to some embodiments, in a case where the air holes **250** extend in a direction inclined with respect to the central axis AXp of the air-hole plate **25**, there is a concern that a region where a flow velocity is relatively small may occur locally in the air-hole plate **25** or at a position relatively close to the air-hole plate **25**. For example, in a case in which hydrogen having a relatively fast combustion speed is used as a fuel, in a case in which backfire occurs, there is a concern that a flame may continuously remain in the above-described region. In a case where the flame continuously remains in the region, there is a concern that the gas turbine combustor **100** may be damaged.

(97) Therefore, in the gas turbine combustor **100** according to some embodiments, damage to the gas turbine combustor **100** is suppressed as follows.

(98) For example, in the gas turbine combustor **100** according to some embodiments, no hydrogen fuel is supplied to the first fuel nozzle **21** when the hydrogen fuel is exclusively combusted.

(99) When the hydrogen fuel is exclusively combusted, no hydrogen fuel is supplied to the first air hole **251** having the inclined passageway **256**, so that it is possible to suppress unintentional continuous remaining of the flame, and it is possible to suppress damage to the gas turbine combustor **100**.

(100) In the gas turbine combustor **100** according to some embodiments, the second air hole group G2 may surround the first air hole group G1 when viewed along the central axis AXp of the air-hole plate **25**.

(101) Since the first air hole **251** has the inclined passageway **256**, a premixed gas of the fuel and the combustion air, which is injected from the first air hole group G1, forms a circulating flow, which makes it easier to retain the flame. In addition, ignition and the flame retention of the

premixed gas injected from the second air hole group G2 can be strengthened by the flame of the premixed gas injected from the first air hole group G1. As a result, a flame retaining property of the gas turbine combustor **100** can be improved while suppressing the continuous remaining of unintended flame.

(102) In the gas turbine combustor **100** according to some embodiments, the first air hole group G1 may be formed in a central region Rc including a position at which the central axis AXp of the air-hole plate **25** passes through the air-hole plate **25**.

(103) As a result, a circulating flow occurs in the premixed gas in a region downstream of the central region Rc, which makes it easier to retain the flame. In addition, ignition and the flame retention of the premixed gas injected from the second air hole group G2 can be strengthened by the flame of the premixed gas injected from the first air hole group G1. As a result, the flame retaining property of the gas turbine combustor **100** can be improved while suppressing the continuous remaining of unintended flame.

(104) In the gas turbine combustor **100** according to the other embodiment, as shown in FIG. 3B, the first air hole group G1 may be formed at a plurality of locations at intervals along a circumferential direction of the air-hole plate **25**.

(105) As a result, a circulating flow occurs in the premixed gas in a region downstream of the plurality of locations at intervals along the circumferential direction of the air-hole plate **25**, which makes it easier to retain the flame. In addition, ignition and the flame retention of the premixed gas injected from the second air hole group G2 can be strengthened by the flame of the premixed gas injected from the first air hole group G1. As a result, the flame retaining property of the gas turbine combustor **100** can be improved while suppressing the continuous remaining of unintended flame.

(106) In the gas turbine combustor **100** according to the other embodiment shown in FIG. 2B and FIG. 3B, the inner peripheral air hole **250i** corresponding to the fuel nozzle **210** of the first row of the central burner **211**, that is, the inner peripheral fuel nozzle **210i**, may be the second air hole **252**.

(107) (About Ratio of Natural Gas Fuel in Premixed Fuel when Natural Gas Fuel is Exclusively Combusted)

(108) In the gas turbine combustor **100** according to some embodiments, the natural gas fuel may be supplied to the plurality of first fuel nozzles **21** and the plurality of second fuel nozzles **22**, such that a ratio of the natural gas fuel in a premixed fuel of the combustion air and the natural gas fuel, which is injected from the first air hole group G1, is larger than a ratio of the natural gas fuel in the premixed fuel, which is injected from the second air hole group G2, when the natural gas fuel is exclusively combusted.

(109) As a result, the flame retaining property when the natural gas fuel is exclusively combusted is improved.

(110) (About Hydrogen Mixed Combustion Ratio During Mixed-Combustion)

(111) In the gas turbine combustor **100** according to some embodiments, the mixed fuel of the hydrogen fuel and the natural gas fuel may be supplied to the plurality of first fuel nozzles **21** and the plurality of second fuel nozzles **22** when a hydrogen mixed combustion ratio, which is a ratio of the hydrogen fuel in the mixed fuel, is equal to or less than a prescribed mixed combustion ratio, and the mixed fuel may be supplied to only the plurality of second fuel nozzles **22** when the hydrogen mixed combustion ratio exceeds the prescribed mixed combustion ratio, when the hydrogen fuel and the natural gas fuel are mixed-combusted.

(112) Since the first air hole **251** corresponding to the first fuel nozzle **21** has the inclined passageway **256**, there is a tendency that the continuous remaining of unintended flame is likely to occur. Therefore, in a case where the hydrogen mixed combustion ratio in the mixed fuel of the hydrogen fuel and the natural gas fuel is relatively high, the continuous remaining of unintended flame is likely to occur, but as the hydrogen mixed combustion ratio decreases, the continuous remaining of unintended flame is less likely to occur.

(113) Therefore, as described above, in a case in which the hydrogen mixed combustion ratio

exceeds the prescribed mixed combustion ratio, the mixed fuel is supplied to only the second fuel nozzle **22**, and no mixed fuel is supplied to the first fuel nozzle **21**, so that the continuous remaining of unintended flame can be suppressed. As a result, the continuous remaining of unintended flame can be suppressed while ensuring the flame retaining property of the gas turbine combustor **100**.

(114) In a case in which the hydrogen mixed combustion ratio can be varied between the fuel supplied to the first fuel nozzle **21** and the fuel supplied to the second fuel nozzle **22** in the fuel system **300**, no hydrogen fuel may be supplied to the first fuel nozzle **21** when the hydrogen fuel and the natural gas fuel are mixed-combusted. In this case, in the fuel system **300**, only the natural gas fuel may be supplyable to the first fuel nozzle **21**.

(115) As a result, it is possible to further suppress the continuous remaining of unintended flame when the hydrogen fuel and the natural gas fuel are mixed-combusted.

(116) In the gas turbine combustor **100** according to some embodiments, the natural gas fuel may be supplied to the first fuel nozzle **21** and the second fuel nozzle **22** when the natural gas fuel is exclusively combusted.

(117) Since the first air hole **251** corresponding to the first fuel nozzle **21** has the inclined passageway **256**, a circulating flow occurs in the premixed gas of the fuel and the combustion air, which is injected from the first air hole **251**, which makes it easier to retain the flame. Therefore, the flame retaining property when the natural gas fuel is exclusively combusted can be improved.

(118) (About Control of Hydrogen Mixed Combustion Ratio)

(119) In the gas turbine combustor **100** according to some embodiments, the hydrogen mixed combustion ratio may be controlled as follows.

(120) FIG. 5A is a view for describing control of combustion injection ratios **Q1**, **Q2**, and **Q3** based on the hydrogen mixed combustion ratio in the gas turbine combustor **100** according to the embodiment shown in FIGS. 2A and 3A.

(121) FIG. 5B is a view for describing control of combustion injection ratios **Q1** and **Q4** based on the hydrogen mixed combustion ratio in the gas turbine combustor **100** according to the other embodiment shown in FIGS. 2B and 3B.

(122) FIG. 5A shows ratios (hereinafter, also referred to as fuel ratios) of the combustion injection ratios **Q1**, **Q2**, and **Q3** with respect to the hydrogen mixed combustion ratio. The combustion injection ratio **Q1** is a combustion injection ratio from the inner peripheral fuel nozzle **210i** of the central burner **211**, the combustion injection ratio **Q2** is a combustion injection ratio from the inner peripheral fuel nozzle **210i** of the outer burner **212**, and the combustion injection ratio **Q3** is a combustion injection ratio from the outer peripheral fuel nozzles **210o** of the central burner **211** and the outer burner **212**.

(123) FIG. 5B shows ratios (fuel ratios) of the combustion injection ratios **Q1** and **Q4** with respect to the hydrogen mixed combustion ratio. The combustion injection ratio **Q1** is the combustion injection ratio from the inner peripheral fuel nozzle **210i** of the central burner **211**, and the combustion injection ratio **Q4** is a combustion injection ratio from the fuel nozzles **210** other than the inner peripheral fuel nozzle **210i** of the central burner **211**.

(124) As shown in FIG. 5A, in the gas turbine combustor **100** according to the embodiment shown in FIGS. 2A and 3A, in a range in which the hydrogen mixed combustion ratio is equal to or more than zero (that is, when the natural gas fuel is exclusively combusted) and equal to or less than a predetermined first mixed combustion ratio **a1** (%), as the hydrogen mixed combustion ratio increases, the combustion injection ratio **Q1** from the inner peripheral fuel nozzle **210i** of the central burner **211** and the combustion injection ratio **Q2** from the inner peripheral fuel nozzle **210i** of the outer burner **212** are gradually decreased, and the combustion injection ratio **Q3** from the outer peripheral fuel nozzles **210o** of the central burner **211** and the outer burner **212** is gradually increased.

(125) In the gas turbine combustor **100** according to the embodiment shown in FIG. 2A and FIG.

3A, in a case in which the hydrogen mixed combustion ratio is equal to or more than the first mixed combustion ratio $a1$ (%), the combustion injection ratio $Q2$ from the inner peripheral fuel nozzle **210i** of the outer burner **212** is a constant value.

(126) In the gas turbine combustor **100** according to the embodiment shown in FIGS. 2A and 3A, in a range in which the hydrogen mixed combustion ratio is equal to or more than the first mixed combustion ratio $a1$ (%) and equal to or less than a predetermined second mixed combustion ratio $a2$ (%), as the hydrogen mixed combustion ratio increases, the combustion injection ratio $Q1$ from the inner peripheral fuel nozzle **210i** of the central burner **211** is gradually decreased, and the combustion injection ratio $Q3$ from the outer peripheral fuel nozzles **210o** of the central burner **211** and the outer burner **212** is gradually increased.

(127) In the gas turbine combustor **100** according to the embodiment shown in FIG. 2A and FIG. 3A, in a case in which the hydrogen mixed combustion ratio is equal to or more than the second mixed combustion ratio $a2$ (%), the combustion injection ratio $Q1$ from the inner peripheral fuel nozzle **210i** of the central burner **211** is zero.

(128) In the gas turbine combustor **100** according to the embodiment shown in FIG. 2A and FIG. 3A, in a case in which the hydrogen mixed combustion ratio is equal to or more than the second mixed combustion ratio $a2$ (%), the combustion injection ratio $Q3$ from the outer peripheral fuel nozzles **210o** of the central burner **211** and the outer burner **212** is a constant value.

(129) In the gas turbine combustor **100** according to the embodiment shown in FIG. 2A and FIG. 3A, in a case in which the hydrogen mixed combustion ratio exceeds the first mixed combustion ratio $a1$ (%), the combustion injection ratio $Q1$ from the inner peripheral fuel nozzle **210i** of the central burner **211** may be zero. That is, the combustion injection ratio $Q1$ from the inner peripheral fuel nozzle **210i** of the central burner **211** may be changed stepwise with the first mixed combustion ratio $a1$ (%) as a boundary.

(130) Similarly, in the gas turbine combustor **100** according to the embodiment shown in FIG. 2A and FIG. 3A, in a case in which the hydrogen mixed combustion ratio exceeds the first mixed combustion ratio $a1$ (%), the combustion injection ratio $Q3$ from the outer peripheral fuel nozzles **210o** of the central burner **211** and the outer burner **212** may be set to the same value as the value in a case in which the hydrogen mixed combustion ratio is equal to or more than the second mixed combustion ratio $a2$ (%) shown in FIG. 5A. That is, the combustion injection ratio $Q3$ from the outer peripheral fuel nozzles **210o** of the central burner **211** and the outer burner **212** may be changed stepwise with the first mixed combustion ratio $a1$ (%) as a boundary.

(131) In the gas turbine combustor **100** according to the embodiment shown in FIGS. 2A and 3A, a rate of change of the combustion injection ratio $Q1$ is changed with the first mixed combustion ratio $a1$ (%) as a boundary, but may remain unchanged.

(132) Similarly, in the gas turbine combustor **100** according to the embodiment shown in FIGS. 2A and 3A, a rate of change of the combustion injection ratio $Q3$ is changed with the first mixed combustion ratio $a1$ (%) as a boundary, but may remain unchanged.

(133) As shown in FIG. 5B, in the gas turbine combustor **100** according to the other embodiment shown in FIGS. 2B and 3B, in a range in which the hydrogen mixed combustion ratio is equal to or more than zero (that is, when the natural gas fuel is exclusively combusted) and equal to or less than the second mixed combustion ratio $a2$ (%), as the hydrogen mixed combustion ratio increases, the combustion injection ratio $Q1$ from the inner peripheral fuel nozzle **210i** of the central burner **211** is gradually decreased, and the combustion injection ratio $Q4$ from the fuel nozzles **210** other than the inner peripheral fuel nozzle **210i** of the central burner **211** is gradually increased.

(134) In the gas turbine combustor **100** according to the other embodiment shown in FIG. 2B and FIG. 3B, in a case in which the hydrogen mixed combustion ratio is equal to or more than the second mixed combustion ratio $a2$ (%), the combustion injection ratio $Q1$ from the inner peripheral fuel nozzle **210i** of the central burner **211** is zero, and all the fuels are injected from the fuel nozzles **210** other than the inner peripheral fuel nozzle **210i** of the central burner **211**.

(135) In the gas turbine combustor **100** according to the other embodiment shown in FIG. 2B and FIG. 3B, in a case in which the hydrogen mixed combustion ratio exceeds the first mixed combustion ratio $a1$ (%), the combustion injection ratio $Q1$ from the inner peripheral fuel nozzle **210i** of the central burner **211** may be zero. That is, the combustion injection ratio $Q1$ from the inner peripheral fuel nozzle **210i** of the central burner **211** may be changed stepwise with the first mixed combustion ratio $a1$ (%) as a boundary.

(136) Similarly, in the gas turbine combustor **100** according to the other embodiment shown in FIG. 2B and FIG. 3B, in a case in which the hydrogen mixed combustion ratio exceeds the first mixed combustion ratio $a1$ (%), the combustion injection ratio $Q4$ from the fuel nozzles **210** other than the inner peripheral fuel nozzle **210i** of the central burner **211** may be set to the same value as the value in a case in which the hydrogen mixed combustion ratio is equal to or more than the second mixed combustion ratio $a2$ (%) shown in FIG. 5B. That is, the combustion injection ratio $Q4$ from the fuel nozzles **210** other than the inner peripheral fuel nozzle **210i** of the central burner **211** may be changed stepwise with the first mixed combustion ratio $a1$ (%) as a boundary.

(137) In the gas turbine combustor **100** according to the other embodiment shown in FIGS. 2B and 3B, the rate of change of the combustion injection ratio $Q1$ is changed with the first mixed combustion ratio $a1$ (%) as a boundary, but may remain unchanged.

(138) Similarly, in the gas turbine combustor **100** according to the other embodiment shown in FIGS. 2B and 3B, a rate of change of the combustion injection ratio $Q4$ is changed with the first mixed combustion ratio $a1$ (%) as a boundary, but may remain unchanged.

(139) (Example of Transition of Hydrogen Mixed Combustion Ratio Over Time)

(140) FIG. 6A is a graph showing a transition of a fuel ratio and the hydrogen mixed combustion ratio from a start of operation of the gas turbine **1** provided with the gas turbine combustor **100** according to the embodiment shown in FIGS. 2A and 3A to a transition to exclusive combustion of the hydrogen fuel.

(141) FIG. 6B is a graph showing a transition of the fuel ratio and the hydrogen mixed combustion ratio from a start of operation of the gas turbine **1** provided with the gas turbine combustor **100** according to the other embodiment shown in FIGS. 2B and 3B to a transition to exclusive combustion of the hydrogen fuel.

(142) As shown in FIGS. 6A and 6B, in the gas turbine combustor **100** according to some embodiments, for example, the operation of the gas turbine **1** may be started by exclusively combusting the natural gas fuel (hydrogen mixed combustion ratio: 0%), and as time elapses, the hydrogen mixed combustion ratio may be gradually increased to transition to the exclusive combustion of the hydrogen fuel (hydrogen mixed combustion ratio: 100%).

(143) (About Control of Fuel Flow Rate Regulating Valves **310** and **320**)

(144) In the gas turbine combustor **100** according to some embodiments shown in FIGS. 2A, 2B, 3A, and 3B, the fuel flow rate control unit **11** controls the fuel flow rate regulating valves **310** and **320** as follows.

(145) (Control of Hydrogen Mixed Combustion Ratio)

(146) In the gas turbine combustor **100** according to some embodiments shown in FIGS. 2A, 2B, 3A, and 3B, the processor **12** of the fuel flow rate control unit **11** calculates opening degrees of the hydrogen flow rate regulating valve **311** and the natural gas flow rate regulating valve **312** such that a fuel having a current hydrogen mixed combustion ratio determined by an operation condition or the like of the gas turbine **1** is generated, for example. The processor **12** outputs the control signal for driving the actuators (not shown) of the hydrogen flow rate regulating valve **311** and the natural gas flow rate regulating valve **312** such that the opening degrees of the hydrogen flow rate regulating valve **311** and the natural gas flow rate regulating valve **312** are the calculated opening degrees.

(147) In the hydrogen flow rate regulating valve **311** and the natural gas flow rate regulating valve **312**, the actuators (not shown) adjust the opening degrees of the hydrogen flow rate regulating

valve **311** and the natural gas flow rate regulating valve **312** by receiving the control signal. As a result, a fuel having a desired hydrogen mixed combustion ratio is generated in the mixing device **307**.

(148) (Control of Fuel Ratio)

(149) In the gas turbine combustor **100** according to some embodiments shown in FIGS. 2A, 2B, 3A, and 3B, the processor **12** of the fuel flow rate control unit **11** calculates opening degrees of the first fuel flow rate regulating valve **321**, the second fuel flow rate regulating valve **322**, the third fuel flow rate regulating valve **323**, and the fourth fuel flow rate regulating valve **324** such that the fuel ratio is a fuel ratio corresponding to the current hydrogen mixed combustion ratio determined by the operation condition or the like of the gas turbine **1**, for example. The processor **12** outputs the control signal for driving the actuator (not shown) of each of the fuel flow rate regulating valves **321**, **322**, **323**, and **324** such that the opening degree of each of the fuel flow rate regulating valves **321**, **322**, **323**, and **324** is the calculated opening degree.

(150) In each of the fuel flow rate regulating valves **321**, **322**, **323**, and **324**, the actuator (not shown) adjusts the opening degree of each of the fuel flow rate regulating valves **321**, **322**, **323**, and **324** by receiving the control signal. As a result, the fuel is injected from each fuel nozzle **210** such that the fuel ratio is the fuel ratio corresponding to the current hydrogen mixed combustion ratio.

(151) The present disclosure is not limited to the above-described embodiments, and includes a modification of the above-described embodiments and an appropriate combination of the embodiments.

(152) For example, the contents described in each embodiment are understood as follows.

(153) (1) A control method for a gas turbine combustor **100** according to at least one embodiment of the present disclosure is a control method for the gas turbine combustor **100** that includes an air-hole plate **25** in which a plurality of air holes **250** are formed and a plurality of fuel nozzles **210** corresponding to the plurality of air holes **250**, respectively, and that combusts a hydrogen fuel and another fuel other than the hydrogen fuel. The plurality of air holes **250** include a plurality of first air holes **251** each having an inclined passageway **256** that extends in a direction inclined with respect to a central axis AXp of the air-hole plate **25** in a region including at least an exit end **250b** in a passageway **255** from an entry end **250a** to the exit end **250b**, and a plurality of second air holes **252** that extend parallel to the central axis AXp. The plurality of fuel nozzles **210** include a plurality of first fuel nozzles **21** corresponding to the plurality of first air holes **251**, respectively, and a plurality of second fuel nozzles **22** corresponding to the plurality of second air holes **252**, respectively. No hydrogen fuel is supplied to the plurality of first fuel nozzles **21** when the hydrogen fuel is exclusively combusted.

(154) According to a method of the above-described (1), when the hydrogen fuel is exclusively combusted, no hydrogen fuel is supplied to the first air hole **251** having the inclined passageway **256**, so that it is possible to suppress continuous remaining of unintended flame, and it is possible to suppress damage to the gas turbine combustor **100**.

(155) (2) In some embodiments, in the method of the above-described (1), the air-hole plate **25** may include a first air hole group G1 in which the plurality of first air holes **251** are disposed adjacent to each other, and a second air hole group G2 in which the plurality of second air holes **252** are disposed adjacent to each other. The second air hole group G2 may surround the first air hole group G1 when viewed along the central axis AXp.

(156) According to a method of the above-described (2), since the first air hole **251** has the inclined passageway **256**, a circulating flow occurs in the premixed gas of the fuel and the combustion air, which is injected from the first air hole group G1, which makes it easier to retain the flame. In addition, ignition and the flame retention of the premixed gas injected from the second air hole group G2 can be strengthened by the flame of the premixed gas injected from the first air hole group G1. As a result, a flame retaining property of the gas turbine combustor **100** can be improved

while suppressing the continuous remaining of unintended flame.

(157) (3) In some embodiments, in the method of the above-described (2), the first air hole group **G1** may be formed in a central region **Rc** including a position at which the central axis **AXp** passes through the air-hole plate **25**.

(158) According to a method of the above-described (3), a circulating flow occurs in the premixed gas in a region downstream of the central region **Rc**, which makes it easier to retain the flame. In addition, ignition and the flame retention of the premixed gas injected from the second air hole group **G2** can be strengthened by the flame of the premixed gas injected from the first air hole group **G1**. As a result, a flame retaining property of the gas turbine combustor **100** can be improved while suppressing the continuous remaining of unintended flame.

(159) (4) In some embodiments, in the method of the above-described (2) or (3), the first air hole group **G1** may be formed at a plurality of locations at intervals along a circumferential direction of the air-hole plate **25**.

(160) According to a method of the above-described (4), a circulating flow occurs in the premixed gas in a region downstream of the plurality of locations at intervals along the circumferential direction of the air-hole plate **25**, which makes it easier to retain the flame. In addition, ignition and the flame retention of the premixed gas injected from the second air hole group **G2** can be strengthened by the flame of the premixed gas injected from the first air hole group **G1**. As a result, a flame retaining property of the gas turbine combustor **100** can be improved while suppressing the continuous remaining of unintended flame.

(161) (5) In some embodiments, in the method of the above-described (3) or (4), the other fuel may be supplied to the plurality of first fuel nozzles **21** and the plurality of second fuel nozzles **22**, such that a ratio of the other fuel in a premixed fuel of combustion air and the other fuel, which is injected from the first air hole group **G1**, is larger than a ratio of the other fuel in the premixed fuel, which is injected from the second air hole group **G2**, when the other fuel is exclusively combusted.

(162) According to a method of the above-described (5), the flame retaining property when the other fuel is exclusively combusted is improved.

(163) (6) In some embodiments, in the method of any one of the above-described (1) to (5), a mixed fuel of the hydrogen fuel and the other fuel may be supplied to the plurality of first fuel nozzles **21** and the plurality of second fuel nozzles **22** when a hydrogen mixed combustion ratio, which is a ratio of the hydrogen fuel in the mixed fuel, is equal to or less than a prescribed mixed combustion ratio, and the mixed fuel may be supplied to only the plurality of second fuel nozzles **22** when the hydrogen mixed combustion ratio exceeds the prescribed mixed combustion ratio, when the hydrogen fuel and the other fuel are mixed-combusted.

(164) According to a method of the above-described (6), even in a case in which the mixed fuel of the hydrogen fuel and the other fuel is injected from the plurality of first fuel nozzles **21**, the continuous remaining of unintended flame can be suppressed. As a result, the continuous remaining of unintended flame can be suppressed while ensuring the flame retaining property of the gas turbine combustor **100**.

(165) (7) In some embodiments, in the method of any one of the above-described (1) to (5), no hydrogen fuel may be supplied to the plurality of first fuel nozzles **21** when the hydrogen fuel and the other fuel are mixed-combusted.

(166) According to a method of the above-described (7), it is possible to suppress the continuous remaining of unintended flame when the hydrogen fuel and the other fuel are mixed-combusted.

(167) (8) In some embodiments, in the method of any one of the above-described (1) to (7), the other fuel may be supplied to the plurality of first fuel nozzles **21** and the plurality of second fuel nozzles **22** when the other fuel is exclusively combusted.

(168) According to a method of the above-described (8), the flame retaining property when the other fuel is exclusively combusted can be improved.

(169) (9) A control device **10** for a gas turbine combustor **100** according to at least one embodiment

of the present disclosure is a control device **10** for controlling combustion in the gas turbine combustor **100** that includes an air-hole plate **25** in which a plurality of air holes **250** are formed and a plurality of fuel nozzles **210** corresponding to the plurality of air holes **250**, respectively, and that combusts a hydrogen fuel and another fuel other than the hydrogen fuel. The plurality of air holes **250** include a plurality of first air holes **251** each having an inclined passageway **256** that extends in a direction inclined with respect to a central axis AXp of the air-hole plate **25** in a region including at least an exit end **250b** in a passageway **255** from an entry end **250a** to the exit end **250b**, and a plurality of second air holes that extend parallel to the central axis. The plurality of fuel nozzles include a plurality of first fuel nozzles **21** corresponding to the plurality of first air holes **251**, respectively, and a plurality of second fuel nozzles **22** corresponding to the plurality of second air holes **252**, respectively. The control device **10** for the gas turbine combustor **100** according to at least one embodiment of the present disclosure includes a fuel flow rate regulating valve **310, 320** that regulates a flow rate of a fuel supplied to the plurality of first fuel nozzles **21**, and a fuel flow rate control unit **11** that controls the fuel flow rate regulating valve **310, 320**. The fuel flow rate control unit **11** controls the fuel flow rate regulating valve **310, 320** such that no hydrogen fuel is supplied to the plurality of first fuel nozzles **21** when the hydrogen fuel is exclusively combusted. (170) According to a configuration of the above-described (9), when the hydrogen fuel is exclusively combusted, no hydrogen fuel is supplied to the first air hole **251** having the inclined passageway **256**, so that it is possible to suppress continuous remaining of unintended flame, and it is possible to suppress damage to the gas turbine combustor **100**.

REFERENCE SIGNS LIST

(171) **10**: control device **11**: fuel flow rate control unit **21**: first fuel nozzle **22**: second fuel nozzle **25**: air-hole plate **100**: gas turbine combustor **210**: fuel nozzle **250**: air hole **250a**: entry end **250b**: exit end **251**: first air hole **252**: second air hole **255**: passageway **256**: inclined passageway **310**: fuel flow rate regulating valve **320**: fuel flow rate regulating valve G1: first air hole group G2: second air hole group Rc: central region

Claims

1. A control method for a gas turbine combustor that includes an air-hole plate in which a plurality of air holes are formed and a plurality of fuel nozzles corresponding to the plurality of air holes, respectively, and that combusts a hydrogen fuel and another fuel other than the hydrogen fuel, in which the plurality of air holes include a plurality of first air holes each having an inclined passageway that extends in a direction inclined with respect to a central axis of the air-hole plate in a region including at least an exit end in a passageway from an entry end to the exit end, and a plurality of second air holes that extend parallel to the central axis, and the plurality of fuel nozzles include a plurality of first fuel nozzles corresponding to the plurality of first air holes, respectively, and a plurality of second fuel nozzles corresponding to the plurality of second air holes, respectively, the control method comprising: supplying the other fuel to the plurality of first fuel nozzles and the plurality of second fuel nozzles when the other fuel is exclusively combusted, and supplying no hydrogen fuel to the plurality of first fuel nozzles when the hydrogen fuel is exclusively combusted.
2. The control method for a gas turbine combustor according to claim 1, wherein the air-hole plate includes: a first air hole group in which the plurality of first air holes are disposed adjacent to each other; and a second air hole group in which the plurality of second air holes are disposed adjacent to each other, and wherein the second air hole group surrounds the first air hole group when viewed along the central axis.
3. The control method for a gas turbine combustor according to claim 2, wherein the first air hole group is formed in a central region including a position at which the central axis passes through the air-hole plate.

4. The control method for a gas turbine combustor according to claim 2, wherein the first air hole group is formed at a plurality of locations at intervals along a circumferential direction of the air-hole plate.

5. The control method for a gas turbine combustor according to claim 3, further comprising: supplying the other fuel to the plurality of first fuel nozzles and the plurality of second fuel nozzles, such that a ratio of the other fuel in a premixed fuel of combustion air and the other fuel, which is injected from the first air hole group, is larger than a ratio of the other fuel in the premixed fuel, which is injected from the second air hole group, when the other fuel is exclusively combusted.

6. The control method for a gas turbine combustor according to claim 1, further comprising: supplying a mixed fuel of the hydrogen fuel and the other fuel to the plurality of first fuel nozzles and the plurality of second fuel nozzles when a hydrogen mixed combustion ratio, which is a ratio of the hydrogen fuel in the mixed fuel is equal to or less than a prescribed mixed combustion ratio and supplying the mixed fuel to only the plurality of second fuel nozzles when the hydrogen mixed combustion ratio exceeds the prescribed mixed combustion ratio, when the hydrogen fuel and the other fuel are mixed-combusted.

7. The control method for a gas turbine combustor according to claim 1, further comprising: supplying no hydrogen fuel to the plurality of first fuel nozzles when the hydrogen fuel and the other fuel are mixed-combusted.

8. A control device for a gas turbine combustor, the control device being for controlling combustion in the gas turbine combustor that includes an air-hole plate in which a plurality of air holes are formed and a plurality of fuel nozzles corresponding to the plurality of air holes, respectively, and that combusts a hydrogen fuel and another fuel other than the hydrogen fuel, in which the plurality of air holes include a plurality of first air holes each having an inclined passageway that extends in a direction inclined with respect to a central axis of the air-hole plate in a region including at least an exit end in a passageway from an entry end to the exit end, and a plurality of second air holes that extend parallel to the central axis, and the plurality of fuel nozzles include a plurality of first fuel nozzles corresponding to the plurality of first air holes, respectively, and a plurality of second fuel nozzles corresponding to the plurality of second air holes, respectively, the control device comprising: a fuel flow rate regulating valve that regulates a flow rate of a fuel supplied to the plurality of first fuel nozzles; and a fuel flow rate control unit that controls the fuel flow rate regulating valve, wherein the fuel flow rate control unit is operable to control the fuel flow rate regulating valve such that the other fuel is supplied to the plurality of first fuel nozzles and the plurality of second fuel nozzles when the other fuel is exclusively combusted, and to control the fuel flow rate regulating valve such that no hydrogen fuel is supplied to the plurality of first fuel nozzles when the hydrogen fuel is exclusively combusted.

9. A control method for a gas turbine combustor that includes an air-hole plate in which a plurality of air holes are formed and a plurality of fuel nozzles corresponding to the plurality of air holes, respectively, and that combusts a hydrogen fuel and another fuel other than the hydrogen fuel, in which the plurality of air holes include a plurality of first air holes each having an inclined passageway that extends in a direction inclined with respect to a central axis of the air-hole plate in a region including at least an exit end in a passageway from an entry end to the exit end, and a plurality of second air holes that extend parallel to the central axis, and the plurality of fuel nozzles include a plurality of first fuel nozzles corresponding to the plurality of first air holes, respectively, and a plurality of second fuel nozzles corresponding to the plurality of second air holes, respectively, the control method comprising: supplying a mixed fuel of the hydrogen fuel and the other fuel to the plurality of first fuel nozzles and the plurality of second fuel nozzles when a hydrogen mixed combustion ratio, which is a ratio of the hydrogen fuel in the mixed fuel is equal to or less than a prescribed mixed combustion ratio and supplying the mixed fuel to only the plurality of second fuel nozzles when the hydrogen mixed combustion ratio exceeds the prescribed mixed combustion ratio, when the hydrogen fuel and the other fuel are mixed-combusted; and

supplying no hydrogen fuel to the plurality of first fuel nozzles when the hydrogen fuel is exclusively combusted.
